Online Appendix C to "Assortment Optimization with Customer-Centric Fairness"

Appendix C.1. Proof for Lemmas and Corollary

Proof of Lemma 2: We have

$$\begin{aligned} R_i^*(C) &= \max_{S:|S|=C} \frac{\sum_{j\in S} r_j v_{ij}}{1+\sum_{j\in S} v_{ij}} \\ &= \max_R\{R|R=\frac{\sum_{j\in S} r_j v_{ij}}{1+\sum_{j\in S} v_{ij}},\ \exists S, |S|=C\} \\ &= \max_R\{R|R=\sum_{j\in S}(r_j-R)v_{ij},\ \exists S, |S|=C\} \\ &= \max_R\{R|R=\max_{S:|S|=C}\sum_{j\in S}(r_j-R)v_{ij}\}, \end{aligned} \tag{A.20}$$

where $\mathcal{S}_i^*(C)$ represents the optimal assortment. Define the linear function $h_{ij}(R)$ to be $h_{ij}(R)=(r_j-R)v_{ij}$. Thus $\mathcal{S}_i^*(C)$ is the optimal solution of $\max_{S:|S|=C}\sum_{j\in S} h_{ij}(R_i^*(C))$, which equals to the summation of the top C largest $h_{ij}(R_i^*(C))$, $j\in\mathcal{N}$. Then we prove the unimodal results by analyzing $C\geq|\mathcal{S}_i^D|$ and $C\leq|\mathcal{S}_i^D|$, respectively.

The case $C\geq|\mathcal{S}_i^D|$: Now we prove that $R_i^*(C)$ decreases when $C\geq|\mathcal{S}_i^D|$. First, we show that when $C>|\mathcal{S}_i^D|$ and for all $j\in\mathcal{N}\backslash\mathcal{S}_i^*(C)$, $h_{ij}(R_i^*(C))\leq 0$. Otherwise, if $j_0\notin\mathcal{S}_i^*(C)$ and $h_{j_0}(R_i^*(C))>0$, it indicates that all the $h_{ij}(R_i^*(C))\geq h_{j_0}(R_i^*(C))>0$ for $j\in\mathcal{S}_i^*(C)$ based on the construction rule of $\mathcal{S}_i^*(C)$. Then

$$\begin{aligned} R_i^*(|\mathcal{S}_i^D|) &\geq R_i^*(C)=\max\nolimits_{S:|S|=C}\textstyle\sum_{j\in S} h_{ij}(R_i^*(C)) \\ &= \max\nolimits_{S:|S|=|\mathcal{S}_i^D|}\textstyle\sum_{j\in S} h_{ij}(R_i^*(C))+\sum_{j\in\mathcal{S}_i^*(C)\backslash\mathcal{S}_i^D} h_{ij}(R_i^*(C)) \\ &> \max\nolimits_{S:|S|=|\mathcal{S}_i^D|}\textstyle\sum_{j\in S} h_{ij}(R_i^*(C)) \\ &\geq \max\nolimits_{S:|S|=|\mathcal{S}_i^D|}\textstyle\sum_{j\in S} h_{ij}(R_i^*(|\mathcal{S}_i^D|))=R_i^*(|\mathcal{S}_i^D|), \end{aligned}$$

where the first inequality is due to the definition of $\mathcal{S}_i^D$; the second inequality is because the second term $\sum_{j\in\mathcal{S}_i^*(C)\backslash\mathcal{S}_i^D} h_{ij}(R_i^*(C))$ is positive; the last inequality is because $h_{ij}(R)$ decreases in R and $R_i^*(C)\leq R_i^*(|\mathcal{S}_i^D|)$. It leads to a contradiction.

Second, we show that when $C=|\mathcal{S}_i^D|$ and for all $j\in\mathcal{N}\backslash\mathcal{S}_i^*(C)$, $h_{ij}(R_i^*(C))\leq 0$. When $C=|\mathcal{S}_i^D|$, the problem is equivalent to the problem without any cardinality constraints, then

$$R_i^*(|\mathcal{S}_i^D|)=\max_S R(S)=\max_R\{R|R=\max_S\sum_{j\in S} h_{ij}(R)\}=\max_R\{R|R=\sum_{j:h_{ij}(R)\geq 0} h_{ij}(R)\}, \tag{A.21}$$

where the second equality is obtained similar to the reformulation process in Equation (A.20). Therefore, we also have $h_{ij}(R_i^*(|\mathcal{S}_i^D|))\leq 0$ for all $j\in\mathcal{N}\backslash\mathcal{S}^*(|\mathcal{S}_i^D|)$.

Therefore, when $C\geq|\mathcal{S}_i^D|$, we have $\max_{k:k\notin\mathcal{S}_i^*(C)} h_k(R_i^*(C))\leq 0$. It leads to

$$R_i^*(C)=\sum_{j\in\mathcal{S}_i^*(C)} h_{ij}(R_i^*(C))\geq\max_{k:k\notin\mathcal{S}_i^*(C)} h_k(R_i^*(C))+\sum_{j\in\mathcal{S}_i^*(C)} h_{ij}(R_i^*(C))=\max_{S:|S|=C+1}\sum_{j\in S} h_{ij}(R_i^*(C)).$$

Note that $\max_{S:|S|=C+1}\sum_{j\in S} h_{ij}(R))$ decreases in R, then $R_i^*(C)\geq R_i^*(C+1)$ otherwise

$$R_i^*(C)<R_i^*(C+1)=\max_{S:|S|=C+1}\sum_{j\in S} h_{ij}(R_i^*(C+1))\leq\max_{S:|S|=C+1}\sum_{j\in S} h_{ij}(R_i^*(C)),$$

which leads to a contradiction. Then we show that $R_i^*(C)\geq R_i^*(C+1)$ when $C\geq|\mathcal{S}_i^D|$ and finish the proof.

The case $C \le |\mathcal{S}_i^D|$: Now we prove that $R_i^*(C)$ increases when $C \le |\mathcal{S}_i^D|$. First, we show that when $C < |\mathcal{S}_i^D|$ and for all $j \in \mathcal{S}_i^*(C)$, $h_{ij}(R_i^*(C)) \ge 0$. Otherwise, if $j_0 \in \mathcal{S}_i^*(C)$ and $h_{j_0}(R_i^*(C)) < 0$, it indicates that all the $h_{ij}(R_i^*(C)) < 0$ for $j \notin \mathcal{S}_i^*(C)$ based on the construction rule of $\mathcal{S}_i^*(C)$. Then

$$\begin{aligned} R_i^*(|\mathcal{S}_i^D|) &\ge R_i^*(C) = \max_{S:|S|=C} h_{ij}(R_i^*(C)) \\ &> \max_{S:|S|=|\mathcal{S}_i^D|} \sum_{j\in S} h_{ij}(R_i^*(C)) \\ &\ge \max_{S:|S|=|\mathcal{S}_i^D|} \sum_{j\in S} h_{ij}(R_i^*(|\mathcal{S}_i^D|)) = R_i^*(|\mathcal{S}_i^D|), \end{aligned}$$

where the second inequality is due to $h_{ij}(R_i^*(C)) < 0$ for all $j \notin \mathcal{S}_i^*(C)$ and the last inequality is because $h_{ij}(R)$ decreases in R. It leads to a contradiction.

Second, we show that when $C = |\mathcal{S}_i^D|$ and for all $j \in \mathcal{S}_i^*(C)$, $h_{ij}(R_i^*(C)) \ge 0$. It is because $R_i^*(|\mathcal{S}_i^D|) = \max_R\{R | R = \sum_{j:h_{ij}(R)\ge 0} h_{ij}(R)\}$ based on Equation (A.21).

Therefore, when $C \le |\mathcal{S}_i^D|$, we have $\min_{k:k\in\mathcal{S}_i^*(C)} h_k(R_i^*(C)) \ge 0$ and thus

$$R_i^*(C) = \sum_{j\in\mathcal{S}_i^*(C)} h_{ij}(R_i^*(C)) \ge \sum_{j\in\mathcal{S}_i^*(C)} h_{ij}(R_i^*(C)) - \min_{k:k\in\mathcal{S}_i^*(C)} h_k(R_i^*(C)) = \max_{S:|S|=C-1} \sum_{j\in S} h_{ij}(R_i^*(C)).$$

Note that $\max_{S:|S|=C-1} \sum_{j\in S} h_{ij}(R))$ decreases in R, then $R_i^*(C) \ge R_i^*(C-1)$ otherwise

$$R_i^*(C) < R_i^*(C-1) = \max_{S:|S|=C-1} \sum_{j\in S} h_{ij}(R_i^*(C-1)) \le \max_{S:|S|=C-1} \sum_{j\in S} h_{ij}(R_i^*(C)),$$

which leads to a contradiction. Then we show that $R_i^*(C-1) \le R_i^*(C)$ when $C \le |\mathcal{S}_i^D|$ and finish the proof. □

Proof of Corollary 1: Based on Theorem 1, if $|\mathcal{S}_2^D| - |\mathcal{S}_1^D| \le \alpha_{2,1}$, Corollary 1 always holds. Then we consider the case $|\mathcal{S}_2^D| - |\mathcal{S}_1^D| > \alpha_{2,1}$, where we have $|\mathcal{S}_1^F| - |\mathcal{S}_2^F| = -\alpha_{2,1}$ and $|\mathcal{S}_2^F| \in [|\mathcal{S}_1^D| + \alpha_{2,1}, |\mathcal{S}_2^D|]$. Note that $\lambda_2 = 1 - \lambda_1$. By the selection rule stated before the corollary, we choose the optimal pair characterized in Theorem 1 with the smallest cardinality for type 1 and type 2.

We first show $|\mathcal{S}_1^F|$ decreases in λ_1. Otherwise, assume there exists λ_1, λ_1' such that $\lambda_1 > \lambda_1'$ and $|\mathcal{S}_1^F(\lambda_1)| > |\mathcal{S}_1^F(\lambda_1')|$, where $\mathcal{S}_i^F(\lambda_1)$ is the optimal solution for group i under the parameter λ_1. Then it indicates $R_1^*(|\mathcal{S}_1^F(\lambda_1)|) \le R_1^*(|\mathcal{S}_1^F(\lambda_1')|)$ and $R_2^*(|\mathcal{S}_1^F(\lambda_1)| + \alpha_{2,1}) \ge R_2^*(|\mathcal{S}_1^F(\lambda_1')| + \alpha_{2,1})$ based on the unimodal property in Lemma 2 and the cardinality range in Theorem 1, where $R_i^*(C)$ is the optimal revenue for the group i when the assortment cardinality is C. Based on Theorem 1, the assortment problem is equivalent to

$$\max_{C\in[|\mathcal{S}_1^D|, |\mathcal{S}_2^D|-\alpha_{2,1}]} \lambda_1 R_1^*(C) + \lambda_2 R_2^*(C + \alpha_{2,1}).$$

Since $\mathcal{S}_i^F(\lambda_1)$ is the optimal assortment for λ_1, $i = 1, 2$. Then we have

$$\lambda_1 R_1^*(|\mathcal{S}_1^F(\lambda_1)|) + (1-\lambda_1) R_2^*(|\mathcal{S}_1^F(\lambda_1)| + \alpha_{2,1}) > \lambda_1 R_1^*(|\mathcal{S}_1^F(\lambda_1')|) + (1-\lambda_1) R_2^*(|\mathcal{S}_1^F(\lambda_1')| + \alpha_{2,1}).$$

Note that the inequality holds strictly based on our solution-selection rule and $|\mathcal{S}_1^F(\lambda_1)| > |\mathcal{S}_1^F(\lambda_1')|$. The above inequality is equivalent to

$$\lambda_1 \left(R_1^*(|\mathcal{S}_1^F(\lambda_1')|) - R_1^*(|\mathcal{S}_1^F(\lambda_1)|)\right) < (1-\lambda_1) \left(R_2^*(|\mathcal{S}_1^F(\lambda_1)| + \alpha_{2,1}) - R_2^*(|\mathcal{S}_1^F(\lambda_1')| + \alpha_{2,1})\right). \quad \text{(A.22)}$$

Similarly, $\mathcal{S}_i^F(\lambda_1')$ is the optimal assortment for λ_1', $i = 1, 2$, indicating

$$\lambda_1' R_1^*(|\mathcal{S}_1^F(\lambda_1')|) + (1-\lambda_1') R_2^*(|\mathcal{S}_1^F(\lambda_1')| + \alpha_{2,1}) \ge \lambda_1' R_1^*(|\mathcal{S}_1^F(\lambda_1)|) + (1-\lambda_1') R_2^*(|\mathcal{S}_1^F(\lambda_1)| + \alpha_{2,1}),$$

which is equivalent to

$$\lambda_1' \left(R_1^*(|\mathcal{S}_1^F(\lambda_1)|) - R_1^*(|\mathcal{S}_1^F(\lambda_1')|)\right) \le (1-\lambda_1') \left(R_2^*(|\mathcal{S}_1^F(\lambda_1')| + \alpha_{2,1}) - R_2^*(|\mathcal{S}_1^F(\lambda_1)| + \alpha_{2,1})\right). \quad \text{(A.23)}$$

Add the left-hand sides and right-hand sides of the inequalities (A.22) and (A.23) to obtain a new inequality:

$$(\lambda_1 - \lambda_1') \left(R_1^*(|\mathcal{S}_1^F(\lambda_1')|) - R_1^*(|\mathcal{S}_1^F(\lambda_1)|)\right) < (\lambda_1' - \lambda_1) \left(R_2^*(|\mathcal{S}_1^F(\lambda_1)| + \alpha_{2,1}) - R_2^*(|\mathcal{S}_1^F(\lambda_1')| + \alpha_{2,1})\right).$$

It leads to a contradiction because the left-hand side is nonnegative while the right-hand side is nonpositive. Therefore, $|\mathcal{S}_1^F|$ decreases in λ_1, which automatically leads to $|\mathcal{S}_1^F|$ and $|\mathcal{S}_2^F|$ decreases in λ_1 (increases in λ_2). □

Proof of Lemma 3. Fix a customer type $i \in \mathcal{M}$. For any cardinality C, recall the reformulation $R_i^*(C) = \max_R \left\{ R : R \le \max_{S\subseteq\mathcal{N}:|S|=C} \sum_{j\in S} h_{ij}(R) \right\}$, where $h_{ij}(R) = v_{ij}(r_j - R)$. For any fixed R, the inner maximization is obtained by choosing the C largest values of $h_{ij}(R)$, $j \in \mathcal{N}$. By Lemma B.3, for each C there exists a threshold index, denoted by $z_i(C)$, such that $\Phi_i(z, C) \ge 0$ for all $z \le z_i(C)$, and $\Phi_i(z, C) < 0$ for all $z > z_i(C)$. Moreover, $R_i^*(C) \in \left[I_i^{(z_i(C))}, I_i^{(z_i(C)+1)} \right)$. Based on previous definition, the order of the linear functions $h_{ij}(R)$, $j \in \mathcal{N}$, is fixed and is represented by $\sigma_i^{(z_i(C))}$ and for every $R \in [I_i^{(z_i(C))}, I_i^{(z_i(C)+1)})$, $\max_{S\subseteq\mathcal{N}:|S|=C} \sum_{j\in S} h_{ij}(R) = \sum_{j\in B_i^{(z_i(C))}(C)} h_{ij}(R)$. By the definition of h_{ij}, the equality above is equivalent to $R_i^*(C) = \frac{\sum_{j\in B_i^{(z_i(C))}(C)} r_j v_{ij}}{1+\sum_{j\in B_i^{(z_i(C))}(C)} v_{ij}}$. Thus, it remains only to show that Algorithm A.3 identifies $z_i(C)$ for every C. Based on the proof of Theorem 2, the position $z_i(C)$ is nondecreasing in C for $C \le D_i$ and nonincreasing for $C \ge D_i$. Therefore, in steps 2 and 3, starting from the previously recorded position, the algorithm correctly records the positions for every C. In Step 2, the first search starts from zero, which is a valid lower bound; in Step 3, the first search starts from z_i^D, which is a valid upper bound. The guarded tests also cover $Z_i = 0$ and the last interval. The values at $C = 0$ and $C = D_i$ are recorded directly in Step 1.

It remains to analyze the running time. For each fixed i, the number of distinct positive intersection coordinates between nonparallel functions $h_{ij}(R)$, $j \in \mathcal{N}$, is at most $O(N^2)$ because two different lines have at most one intersection. Thus, the interval construction and product reordering in Step 1 take $O(N^2 \log N)$ time per type, including the sorting of the intersection records. In Step 2, the index z only increases, so the total number of upward movements over all $C < D_i$ is at most Z_i. In Step 3, the index z only decreases, so the total number of downward movements over all $C > D_i$ is also at most Z_i. Thus, for each customer type, the total number of interval-position updates is $O(Z_i) = O(N^2)$. Therefore, the total running time, including interval construction and product reordering for all M types, is $O\big(M\big(N^2 \log N\big)\big) = O(MN^2 \log N)$. This completes the proof. □

Proof of Lemma 4. Among all optimal solutions, choose $\boldsymbol{C}^F$ to minimize $L(\boldsymbol{C}) := \sum_{i\in\mathcal{M}} |C_i - D_i|$. Suppose that some $C_i^F \notin [|\mathcal{S}_1^D|, |\mathcal{S}_M^D|]$, and define $\widehat{C}_i := \min\{|\mathcal{S}_M^D|, \max\{|\mathcal{S}_1^D|, C_i^F\}\}$. Since truncation to an interval is nondecreasing and 1-Lipschitz, for every $i_1, i_2 \in \mathcal{M}$, $i_1 \ne i_2$,

$$\widehat{C}_{i_1} - \widehat{C}_{i_2} \le \max\{C_{i_1}^F - C_{i_2}^F, 0\} \le \alpha_{i_1,i_2}.$$

Hence, $\widehat{\boldsymbol{C}}$ is feasible. Moreover, truncation moves each C_i^F weakly toward $|\mathcal{S}_i^D|$, so Lemma 2 implies $R_i^*(\widehat{C}_i) \ge R_i^*(C_i^F)$ for every i. Thus, $\widehat{\boldsymbol{C}}$ is also optimal, but $L(\widehat{\boldsymbol{C}}) < L(\boldsymbol{C}^F)$, contradicting the choice of $\boldsymbol{C}^F$. Therefore, $|\mathcal{S}_1^D| \le C_i^F \le |\mathcal{S}_M^D|$ for every $i \in \mathcal{M}$. □

Proof of Lemma 5. We first restrict the cardinality range without changing the optimal value. By Lemma 4, adding the bounds $D_1 \le C_i \le D_M$ for all $i \in \mathcal{M}$ to Problem (3) does not change its optimal value. It therefore suffices to establish a correspondence between this restricted problem and Problem (4)-(7).

First, we show that every feasible binary solution yields a feasible cardinality vector with the same objective value. Given any feasible solution $\boldsymbol{y}$ to Problem (4)-(7), define $C_i = D_1 + \sum_{k=D_1+1}^{D_M} y_{i,k}$, $i \in \mathcal{M}$. By Constraint (5), the entries $y_{i,k}$ for each type consist of consecutive ones followed by zeros. Hence, $C_i \in \{D_1, \ldots, D_M\}$ and $y_{i,k} = \mathbf{1}_{C_i \ge k}$ for every defined level k. Suppose that $C_{i_1} - C_{i_2} > \alpha_{i_1,i_2}$ for some $i_1, i_2 \in \mathcal{M}$, $i_1 \ne i_2$ for a contradiction. Define $k = C_{i_1}$, then $y_{i_1,k} = 1$ and $y_{i_2,k-\alpha_{i_1,i_2}} = 0$ because $k - \alpha_{i_1,i_2} > C_{i_2}$. This contradicts Constraint (6). Therefore, $\boldsymbol{C}$ is feasible for the restricted problem and hence for Problem (3). The construction also preserves the objective value. By the definition of $w_{i,k}$, for every $i \in \mathcal{M}$,

$$\lambda_i R_i^*(D_1) + \sum_{k=D_1+1}^{D_M} w_{i,k} y_{i,k} = \lambda_i R_i^*(D_1) + \lambda_i \sum_{k=D_1+1}^{C_i} \big(R_i^*(k) - R_i^*(k-1)\big) = \lambda_i R_i^*(C_i).$$

Summing over all customer types shows that $\boldsymbol{C}$ attains the same objective value in Problem (3) as $\boldsymbol{y}$ does in Problem (4)-(7).

Conversely, every feasible cardinality vector in the restricted problem yields a feasible binary solution with the same objective value. Given any feasible solution $\boldsymbol{C}$ to Problem (3) satisfying $D_1 \le C_i \le D_M$ for all $i \in \mathcal{M}$, define

$$y_{i,k} = 1\{C_i \ge k\}, \qquad i \in \mathcal{M}, \quad k = D_1 + 1, \ldots, D_M.$$

These variables satisfy Constraints (5) and (7). For every index appearing in Constraint (6), if $y_{i_1,k} = 0$, the inequality holds automatically. Otherwise, $C_{i_1} \geq k$, and the feasibility of $\boldsymbol{C}$ implies

$$C_{i_2} \geq C_{i_1} - \alpha_{i_1,i_2} \geq k - \alpha_{i_1,i_2}.$$

Consequently, $y_{i_2,k-\alpha_{i_1,i_2}} = 1$, so Constraint (6) also holds. Thus, $\boldsymbol{y}$ is feasible for Problem (4)-(7). The same telescoping identity above shows that $\boldsymbol{y}$ attains the same objective value as $\boldsymbol{C}$.

The two constructions establish that the restricted problem and Problem (4)-(7) have the same optimal value. Since the restriction does not change the optimal value of Problem (3), the claimed equivalence follows. Moreover, the first construction recovers an optimal solution to Problem (3) from any optimal solution to Problem (4)-(7). □

Proof of Lemma 6: First, for any pair in the support of an optimal policy, write $C_i = |\mathcal{S}_i|$ and replace $\mathcal{S}_i$ by $\mathcal{S}_i^*(\widehat{C}_i)$, where $\widehat{C}_i = \min\{|\mathcal{S}_2^D|, \max\{|\mathcal{S}_1^D|, C_i\}\}$, $i = 1, 2$. Lemma 2 implies that this replacement does not decrease revenue, and $(\widehat{C}_i - \widehat{C}_j - \alpha_{i,j})^+ \leq (C_i - C_j - \alpha_{i,j})^+$, $i, j \in \{1, 2\}$, so both fairness constraints remain satisfied. Thus, we can restrict all support pairs to $|\mathcal{S}_1^D| \leq |\mathcal{S}_i| \leq |\mathcal{S}_2^D|$, $i = 1, 2$, without losing optimality.

Assume that there exists an optimal strategy G_0 such that $P_{G_0}((\mathcal{S}_1^0, \mathcal{S}_2^0)) > 0$ and $|\mathcal{S}_1^0| > |\mathcal{S}_2^0|$. Let $\boldsymbol{\mathcal{S}}^0 = (\mathcal{S}_1^0, \mathcal{S}_2^0)$ and $\widehat{\boldsymbol{S}} = (\mathcal{S}_1^0, \mathcal{S}_2^*(|\mathcal{S}_1^0|))$. These pairs are distinct because their second assortments have different cardinalities. Construct G' by setting

$$P_{G'}(\boldsymbol{S}) = \begin{cases} 0, & \boldsymbol{S} = \boldsymbol{\mathcal{S}}^0, \\ P_{G_0}(\widehat{\boldsymbol{S}}) + P_{G_0}(\boldsymbol{\mathcal{S}}^0), & \boldsymbol{S} = \widehat{\boldsymbol{S}}, \\ P_{G_0}(\boldsymbol{S}), & \text{otherwise.} \end{cases}$$

Here, $\mathcal{S}_2^*(|\mathcal{S}_1^0|)$ is an optimal assortment for group 2 with cardinality $|\mathcal{S}_1^0|$. We will show G' is feasible and will not decrease the revenue. First, the probability is well-defined and satisfies the fairness constraints since $\mathbb{E}_{G'}\left[(|\mathcal{S}_2| - |\mathcal{S}_1| - \alpha_{2,1})^+\right] = \mathbb{E}_{G_0}\left[(|\mathcal{S}_2| - |\mathcal{S}_1| - \alpha_{2,1})^+\right] \leq \kappa_1$ and $\mathbb{E}_{G'}\left[(|\mathcal{S}_1| - |\mathcal{S}_2| - \alpha_{1,2})^+\right] \leq \mathbb{E}_{G_0}\left[(|\mathcal{S}_1| - |\mathcal{S}_2| - \alpha_{1,2})^+\right] \leq \kappa_2$. Second, by the initial cardinality restriction, we have $|\mathcal{S}_1^0| \leq |\mathcal{S}_2^D|$ so the revenue for group 2 will not decrease when increasing the cardinality of $\mathcal{S}_2$.

We can repeat the process until we construct a strategy such that $P_G((\mathcal{S}_1, \mathcal{S}_2)) = 0$ for every $|\mathcal{S}_1| > |\mathcal{S}_2|$, and the revenue is no less than the maximal revenue brought by G_0. Then we finish the proof. □

Proof of Lemma 7. For every $i \in \mathcal{M}$ and $k = |\mathcal{S}_1^D| + 1, ..., |\mathcal{S}_M^D|$, define the binary variable $y_{i,k} = \mathbf{1}\{C_i \geq k\}$ and the marginal revenue weight $w_{i,k} = \lambda_i(R_i^*(k) - R_i^*(k-1))$. Then, for every $\boldsymbol{C} \in \mathcal{D}^M$, $F(\boldsymbol{C})$ can be represented by

$$F(\boldsymbol{C}) = \sum_{i \in \mathcal{M}} \lambda_i R_i^*(|\mathcal{S}_1^D|) + \sum_{i \in \mathcal{M}} \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} w_{i,k} y_{i,k}.$$

Moreover, $U_i(\boldsymbol{C})$ can also be expressed by $\boldsymbol{y}$. Specifically, for any ordered pair (m, i), we have

$$(C_m - C_i - \alpha_{m,i})^+ = \sum_{k=\alpha_{m,i}+|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} \mathbf{1}\{C_m \geq k\}\mathbf{1}\{C_i < k - \alpha_{m,i}\} = \sum_{k=\alpha_{m,i}+|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{m,k}\left(1 - y_{i,k-\alpha_{m,i}}\right).$$

The sum is interpreted as zero when $\alpha_{m,i} + |\mathcal{S}_1^D| + 1 > |\mathcal{S}_M^D|$. Therefore, the objective function of $\Phi(\boldsymbol{\mu})$ can be represented by

$$\sum_{i \in \mathcal{M}} \lambda_i R_i^*(|\mathcal{S}_1^D|) + \sum_{i \in \mathcal{M}} \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} w_{i,k} y_{i,k} - \sum_{i \in \mathcal{M}} \mu_i \sum_{m \in \mathcal{M},\ m \neq i} \sum_{k=\alpha_{m,i}+|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{m,k}\left(1 - y_{i,k-\alpha_{m,i}}\right)$$

The consecutive relation results from the similar analysis of Constraint (5). Conversely, any binary vector satisfying these consecutive relations defines $C_i = |\mathcal{S}_1^D| + \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{i,k}$ in $\mathcal{D}^M$, and its entries satisfy $y_{i,k} = \mathbf{1}\{C_i \geq k\}$. The same identities then preserve the objective value, establishing equivalence in both directions. □

Appendix C.2. Proofs of Section 6

Proof of Lemma 8: We only need to prove that for any $C_1 \le C_2$, we have $\frac{R_i^*(C_1)}{C_1} \ge \frac{R_i^*(C_2)}{C_2}$. Recall that

$$R_i^*(C) = \max_{S:|S|=C} \frac{\sum_{j\in S} r_j v_{ij}}{1+\sum_{j\in S} v_{ij}} = \max_R R \text{ s.t. } \max_{S:|S|=C} \sum_{j\in S} h_{ij}(R) \ge R,$$

where $h_{ij}(R) = (r_j - R)v_{ij}$ and $\max_{S:|S|=C} \sum_{j\in S} h_{ij}(R)$ is equivalent to the summation of the top C value of $h_{ij}(R)$, $j \in \mathcal{N}$ for each R. Denote the Cth largest $h_{ij}(R)$ under a fixed R as $h_j^{(C)}(R)$.

For any $R \ge 0$, we have

$$\begin{aligned}
\max_{S:|S|=C_2} \sum_{j\in S} h_{ij}(R) &\le \max_{S:|S|=C_1} \sum_{j\in S} h_{ij}(R) + (C_2 - C_1) h_j^{(C_1)}(R) \\
&\le \max_{S:|S|=C_1} \sum_{j\in S} h_{ij}(R) + \tfrac{(C_2-C_1)}{C_1} C_1 h_j^{(C_1)}(R) \\
&\le \max_{S:|S|=C_1} \sum_{j\in S} h_{ij}(R) + \tfrac{(C_2-C_1)}{C_1} \max_{S:|S|=C_1} \sum_j h_j^{(C_1)}(R) \\
&= \tfrac{C_2}{C_1} \max_{S:|S|=C_1} \sum_j h_j^{(C_1)}(R),
\end{aligned}$$

where the first and third inequality are because $h_j^{(C)}(R)$ decreases in C and $C_1 \le C_2$. It leads to the following inequality

$$\begin{aligned}
R_i^*(C_2) &= \max_R R \text{ s.t. } \max_{S:|S|=C_2} \sum_{j\in S} (r_j - R) v_{ij} \ge R \\
&\le \max_R R \text{ s.t. } \tfrac{C_2}{C_1} \max_{S:|S|=C_1} \sum_{j\in S} h_{ij}(R) \ge R.
\end{aligned}$$

Set the variable $x = \frac{C_1}{C_2} R$. Then we have

$$\begin{aligned}
R_i^*(C_2) &\le \tfrac{C_2}{C_1} \max_x x \text{ s.t. } \tfrac{C_2}{C_1} \max_{S:|S|=C_1} \sum_{j\in S} h_{ij}(\tfrac{C_2}{C_1} x) \ge \tfrac{C_2}{C_1} x \\
&= \tfrac{C_2}{C_1} \max_x x \text{ s.t. } \max_{S:|S|=C_1} \sum_{j\in S} h_{ij}(\tfrac{C_2}{C_1} x) - x \ge 0 \\
&\le \tfrac{C_2}{C_1} \max_x x \text{ s.t. } \max_{S:|S|=C_1} \sum_{j\in S} h_{ij}(x) - x \ge 0 \\
&= \tfrac{C_2}{C_1} R_i^*(C_1),
\end{aligned}$$

where the last inequality is because $h_{ij}(x)$ decreases in x for all j. This completes the proof. □

Proof of Proposition 3. (i) The first bound: Select $h \in \arg\max_i \lambda_i R_i^*(|\mathcal{S}_i^D|)$. Set $\mathcal{S}_i^F = \mathcal{S}_i^*(|\mathcal{S}_h^D|)$ for all $i \in \mathcal{M}$. Then the assortment is feasible and we have $\Pi_S^F/\Pi^D \ge 1/M$.

(ii) The second bound: Recall that $D_i^U = \min_{j\in\mathcal{M}}\{D_j + d_{i,j}\}$, where $D_i = |\mathcal{S}_i^D|$. Since $d_{i,j} \ge 0$, $d_{i,i} = 0$, and $D_j \ge D_1$, we have $D_1 \le D_i^U \le D_i$ for every $i \in \mathcal{M}$.

We first show that the assortment with the cardinality vector $\boldsymbol{C} = \boldsymbol{D}^U$ is feasible for the original fairness constraints. Fix any distinct $i, j \in \mathcal{M}$. The triangle inequality for the shortest-path distances gives

$$D_i^U = \min_{m\in\mathcal{M}}\{D_m + d_{i,m}\} \le \min_{m\in\mathcal{M}}\{D_m + d_{j,m} + d_{i,j}\} = D_j^U + d_{i,j} \le D_j^U + \alpha_{i,j}.$$

which implies $D_i^{\mathrm{U}} - D_j^{\mathrm{U}} \le \alpha'_{i,j}$. Therefore, such assortment decision with cardinality $\boldsymbol{D}^{\mathrm{U}}$ is feasible.

Therefore, the optimal revenue satisfies $\Pi_S^F \ge \sum_{i\in\mathcal{M}} \lambda_i R_i^*(D_i^{\mathrm{U}})$. Note that Lemma 8 implies $R_i^*(D_i^{\mathrm{U}})/D_i^{\mathrm{U}} \ge R_i^*(|\mathcal{S}_i^D|)/|\mathcal{S}_i^D|$. Therefore,

$$\Pi_S^F \ge \sum_{i\in\mathcal{M}} \lambda_i R_i^*(D_i) \frac{D_i^{\mathrm{U}}}{|\mathcal{S}_i^D|}.$$

Dividing by Π^D gives

$$\frac{\Pi_S^F}{\Pi^D} \ge \sum_{i\in\mathcal{M}} \frac{\lambda_i R_i^*(D_i)}{\Pi^D} \frac{D_i^{\mathrm{U}}}{|\mathcal{S}_i^D|} \ge \min_{i\in\mathcal{M}} \frac{D_i^{\mathrm{U}}}{|\mathcal{S}_i^D|},$$

where the last inequality is because $\Pi^D = \sum_{i\in\mathcal{M}} \lambda_i R_i^*(D_i)$.

(iii) The last bound: Let $\omega_i = \lambda_i R_i^*(|\mathcal{S}_i^D|)/\Pi^D, \quad i \in \mathcal{M}$. Thus, ω_i is type i's share of the fully customized revenue, and $\sum_{i\in\mathcal{M}} \omega_i = 1$. Clearly, $\omega_i \le \Pi_S^F/\Pi^D$ for every $i \in \mathcal{M}$. Recall from part (ii) that $\boldsymbol{D}^U = (D_i^U)_{i\in\mathcal{M}}$ is a feasible cardinality vector and $R_i^*(D_i^U)/R_i^*(|\mathcal{S}_i^D|) \ge D_i^U/|\mathcal{S}_i^D|$. Consequently,

$$\frac{\Pi_S^F}{\Pi^D} \ge \frac{\sum_{i\in\mathcal{M}} \lambda_i R_i^*(D_i^U)}{\Pi^D} \ge \sum_{i\in\mathcal{M}} \omega_i \frac{D_i^U}{|\mathcal{S}_i^D|} = 1 - \sum_{i\in\mathcal{M}} \omega_i \left(1 - \frac{D_i^U}{|\mathcal{S}_i^D|}\right) \ge 1 - \frac{\Pi_S^F}{\Pi^D} \sum_{i\in\mathcal{M}} \left(1 - \frac{D_i^U}{|\mathcal{S}_i^D|}\right).$$

Rearranging the last inequality yields

$$\frac{\Pi_S^F}{\Pi^D} \ge \frac{1}{1 + \sum_{i\in\mathcal{M}} \left(1 - \frac{D_i^U}{|\mathcal{S}_i^D|}\right)}.$$

In addition, if the fully customized assortment satisfies all fairness constraints, then $D_i \le D_j + d_{i,j}$ for every $i, j \in \mathcal{M}$. Since $d_{i,i} = 0$, this gives $D_i^U = D_i = |\mathcal{S}_i^D|$ for every $i \in \mathcal{M}$. Hence the second term of our lower bound will become 1, which leads to $L_S = 1$. □

Proof of Proposition 4. Let $(\mathcal{S}_i^F)_{i\in\mathcal{M}}$ be an optimal deterministic assortment strategy for Problem (DFA-M) with tolerance level $\boldsymbol{\alpha}$. Consider the randomized policy that offers $(\mathcal{S}_i^D)_{i\in\mathcal{M}}$ with probability p^0 and $(\mathcal{S}_i^F)_{i\in\mathcal{M}}$ with probability $1 - p^0$.

Because $(\mathcal{S}_i^F)_{i\in\mathcal{M}}$ has zero excess unfairness, the expected unfairness perceived by type i under this policy is

$$p^0 \sum_{m\in\mathcal{M}} \left(|\mathcal{S}_m^D| - |\mathcal{S}_i^D| - \alpha_{m,i}\right)^+ \le \kappa_i,$$

where the inequality follows from the definition of p^0. Hence the policy is feasible for Problem (RFA-M). Its expected revenue is $p^0\Pi^D + (1-p^0)\Pi_S^F$. Therefore, the optimality of G^F and Proposition 3 imply

$$\frac{\Pi_G^F}{\Pi^D} \ge p^0 + (1-p^0)\frac{\Pi_S^F}{\Pi^D} \ge p^0 + (1-p^0)L_S.$$

If the fully customized policy satisfies all randomized fairness constraints, then $\sum_{m\in\mathcal{M}} \left(|\mathcal{S}_m^D| - |\mathcal{S}_i^D| - \alpha_{m,i}\right)^+ \le \kappa_i$ for every $i \in \mathcal{M}$, so $p^0 = 1$ and the lower bound will become 1. □

Proof of Proposition 5. (i) Lower bound: The feasible policy constructed in the proof of Proposition 4 gives $\Pi_G^F \ge p^0\Pi^D + (1-p^0)\Pi_S^F$. Dividing both sides by Π_S^F yields

$$\frac{\Pi_G^F}{\Pi_S^F} \ge 1 + p^0 \left(\frac{\Pi^D}{\Pi_S^F} - 1\right).$$

(ii) Upper bound: First, no randomized policy can earn more than the fully customized policy, so $\Pi_G^F \le \Pi^D$, and hence $\Pi_G^F/\Pi_S^F \le \Pi^D/\Pi_S^F$. Here, we use the shortest-path distances $d_{i,j}$ from Appendix A.2 to prove the result.

Suppose we select an optimal randomized policy such that every cardinality vector $\boldsymbol{C}$ in its support belongs to $\mathcal{D}^M$. Following the definition $D_i^U = \min_{j\in\mathcal{M}}\{|\mathcal{S}_j^D| + d_{i,j}\}$, for every such $\boldsymbol{C}$, define

$$C_i^U := \min_{j\in\mathcal{M}}\{C_j + d_{i,j}\} \in [|\mathcal{S}_1^D|, C_i], \qquad i \in \mathcal{M}.$$

Similar to the proof in Proposition 3, $\boldsymbol{C}^U = (C_i^U)_{i\in\mathcal{M}}$ is feasible for deterministic fairness problem, that is $C_i^U - C_j^U \le d_{i,j}$ for every $i, j \in \mathcal{M}$.

We next bound the reduction from C_i to C_i^U. If $C_i = C_i^U$, the required bound is immediate. Otherwise, suppose $C_i^U = C_{j^\star} + d_{i,j^\star}$, and let $i = i_0, i_1, \ldots, i_L = j^\star$ be a simple shortest path attaining $d_{i,j^\star}$. Since $d_{i,j^\star} < C_i - C_{j^\star} \le N$, every arc on this path has length equal to its original tolerance. Then

$$C_i - C_i^U = C_i - C_{j^\star} - d_{i,j^\star} = \sum_{\ell=0}^{L-1} \left(C_{i_\ell} - C_{i_{\ell+1}} - \alpha_{i_\ell,i_{\ell+1}}\right) \le \sum_{\ell=0}^{L-1} \left(C_{i_\ell} - C_{i_{\ell+1}} - \alpha_{i_\ell,i_{\ell+1}}\right)^+ \le \sum_{i\in\mathcal{M}} U_i(\boldsymbol{C}).$$

Since $C_i^U \le C_i$, Lemma 8 implies

$$R_i^*(C_i) \le \frac{C_i}{C_i^U} R_i^*(C_i^U) = \left(1 + \frac{C_i - C_i^U}{C_i^U}\right) R_i^*(C_i^U) \le \left(1 + \frac{\sum_{q\in\mathcal{M}} U_q(C)}{|\mathcal{S}_1^D|}\right) R_i^*(C_i^U).$$

After multiplying by λ_i and summing over i, we obtain

$$\sum_i \lambda_i R_i^*(C_i) \le \left(1 + \frac{\sum_{q\in\mathcal{M}} U_q(C)}{|\mathcal{S}_1^D|}\right) \sum_i \lambda_i R_i^*(C_i^U) \le \left(1 + \frac{\sum_{q\in\mathcal{M}} U_q(C)}{|\mathcal{S}_1^D|}\right) \Pi_{\mathcal{S}}^F,$$

where the last inequality is due to the feasibility of $\boldsymbol{C}^U$ to the deterministic problem. Taking expectations under G^F and applying the randomized fairness constraints yields

$$\Pi_G^F = \mathbb{E}_{G^F}[\sum_i \lambda_i R_i^*(C_i)] \le \left(1 + \frac{\sum_{i\in\mathcal{M}} \mathbb{E}_{G^F}[U_i(C)]}{|\mathcal{S}_1^D|}\right) \Pi_{\mathcal{S}}^F \le \left(1 + \frac{\sum_{q\in\mathcal{M}} \kappa_q}{|\mathcal{S}_1^D|}\right) \Pi_{\mathcal{S}}^F.$$

Dividing by $\Pi_{\mathcal{S}}^F$ proves the upper bound. □

Proof of Proposition 6. By Proposition 1, choose an optimal policy G^F that offers fixed-cardinality optimal assortments $\mathcal{S}_i^*(C_i)$. Fix a customer type i such that $C_i \ge |\mathcal{S}_i^D|$ in the support of G^F. We first show that $W_i(\mathcal{S}_i^*(C_i)) \ge W_i(\mathcal{S}_i^D)$. If $\mathcal{S}_i^D \subseteq \mathcal{S}_i^*(C_i)$, then $W_i(\mathcal{S}_i^*(C_i)) \ge W_i(\mathcal{S}_i^D)$ automatically holds. Otherwise, assume $\mathcal{S}_i^D \backslash \mathcal{S}_i^*(C_i) \ne \emptyset$. If $R_i(\mathcal{S}_i^*(C_i)) = R_i(\mathcal{S}_i^D)$, then $\mathcal{S}_i^*(C_i)$ is also fully customized optimal and must contain every product with $r_j > R_i(\mathcal{S}_i^D)$: adding any missing such product would strictly increase revenue. By our smallest-cardinality convention, $\mathcal{S}_i^D = \{j : r_j > R_i(\mathcal{S}_i^D)\}$, so equality would imply $\mathcal{S}_i^D \subseteq \mathcal{S}_i^*(C_i)$, a contradiction. Thus, in the case under consideration, $R_i(\mathcal{S}_i^*(C_i)) < R_i(\mathcal{S}_i^D)$. Recall that $\mathcal{S}_i^D = \arg\max_{S:|S|=|\mathcal{S}_i^D|} \sum_{j\in S} h_{ij}(R_i(\mathcal{S}_i^D))$, $\mathcal{S}_i^*(C_i) = \arg\max_{S:|S|=|\mathcal{S}_i^*(C_i)|} \sum_{j\in S} h_{ij}(R_i(\mathcal{S}_i^*(C_i)))$ and $R_i(\mathcal{S}_i^*(C_i)) < R_i(\mathcal{S}_i^D)$. Therefore, for any $j_1 \in \mathcal{S}_i^*(C_i)\backslash\mathcal{S}_i^D$ and $j_2 \in \mathcal{S}_i^D\backslash\mathcal{S}_i^*(C_i)$, we have $h_{ij_1}(R_i(\mathcal{S}_i^*(C_i))) \ge h_{ij_2}(R_i(\mathcal{S}_i^*(C_i)))$ and $h_{ij_1}(R_i(\mathcal{S}_i^D)) \le h_{ij_2}(R_i(\mathcal{S}_i^D))$. Since $h_{ij}(R) = v_{ij}(r_j - R)$ is a linear function in terms of R and $R_i(\mathcal{S}_i^*(C_i)) < R_i(\mathcal{S}_i^D)$, then we have $v_{ij_1} \ge v_{ij_2}$ for any $j_1 \in \mathcal{S}_i^*(C_i)\backslash\mathcal{S}_i^D$ and $j_2 \in \mathcal{S}_i^D\backslash\mathcal{S}_i^*(C_i)$. In addition to the fact $|\mathcal{S}_i^*(C_i)| \ge |\mathcal{S}_i^D|$, it follows that $\sum_{j\in\mathcal{S}_i^*(C_i)} v_{ij} \ge \sum_{j\in\mathcal{S}_i^D} v_{ij}$ and $W_i(\mathcal{S}_i^*(C_i)) \ge W_i(\mathcal{S}_i^D)$.

Since this inequality holds for every $\boldsymbol{C}$ in the support of G^F, taking expectation yields $\mathbb{E}_{G^F}[W_i(\mathcal{S}_i)] \ge W_i(\mathcal{S}_i^D)$.

Deterministic special case. Finally, supposing that $\kappa_i = 0$ for every $i \in \mathcal{M}$, the randomized problem reduces to the deterministic fairness problem. Similar to the above proof, for any type i satisfying $|\mathcal{S}_i^F| \ge |\mathcal{S}_i^D|$, we have $W_i(\mathcal{S}_i^F) \ge W_i(\mathcal{S}_i^D)$. □

Appendix C.3. Proof of the Preprocessing Result in the Appendix

Proof of Lemma A.1: We first show that Step 2 preserves the feasible region and produces tight bounds. Capping these tolerances at N does not change $\mathcal{F}_0$. Suppose we have a feasible solution under d_{i_1,i_2} to be $\mathcal{S}_i'$, $i \in \mathcal{M}$. Based on the definition, we have $|\mathcal{S}_{i_1}'| - |\mathcal{S}_{i_2}'| \le d_{i_1,i_2} \le \alpha_{i_1,i_2}$ for any $i_1, i_2 \in \mathcal{M}$. So $\mathcal{S}_i'$ is also feasible for Problem (DFA-M). On the other hand, suppose we have a feasible solution under α_{i_1,i_2} to be $\mathcal{S}_i$, $i \in \mathcal{M}$. Assume d_{i_1,i_2} is obtained by the shortest path $i_{k_1} \to i_{k_2} \to \cdots \to i_{k_K}$, where $i_{k_1} = i_1$, $i_{k_K} = i_2$ and $d_{i_1,i_2} = \sum_{l\le K-1} \min\{N, \alpha_{i_{k_l},i_{k_{l+1}}}\}$. Then for any $i_1, i_2 \in \mathcal{M}$, we have

$$|\mathcal{S}_{i_1}| - |\mathcal{S}_{i_2}| = \sum_{l\le K-1} \left(|\mathcal{S}_{i_{k_l}}| - |\mathcal{S}_{i_{k_{l+1}}}|\right) \le \sum_{l\le K-1} \min\{N, \alpha_{i_{k_l},i_{k_{l+1}}}\} = d_{i_1,i_2}.$$

Thus $\mathcal{S}_i$, $i \in \mathcal{M}$ is also feasible for the problem under d_{i_1,i_2} and the two problems are equivalent. Moreover, fix $j \in \mathcal{M}$ and set $C_k = d_{k,j}$ for every $k \in \mathcal{M}$. These cardinalities are integers in $[0, N]$, and the triangle inequality gives $C_k - C_\ell = d_{k,j} - d_{\ell,j} \le d_{k,\ell} \le \alpha_{k,\ell}$ for every distinct k, ℓ. Thus, this constructed vector $\boldsymbol{C}$ belongs to $\mathcal{F}_0$ and attains $C_i - C_j = d_{i,j}$ for every i because $d_{j,j} = 0$. Consequently,

$$d_{i,j} = \max_{\boldsymbol{C}\in\mathcal{F}_0} (C_i - C_j), \qquad i, j \in \mathcal{M}. \tag{C.24}$$

We next show that the constraints retained in Step 3 preserve $\mathcal{F}_0$. We prove that every vector satisfying the retained constraints belongs to $\mathcal{F}_0$, and conversely. Let E denote the ordered class pairs retained in Step 3.2. The class distance $d_{Q,R}$ is independent of the chosen representatives by the triangle inequality and the zero distances in both directions within each class. Take any vector $\boldsymbol{C} \in \{0, 1, \ldots, N\}^M$ satisfying the retained constraints. The cycles in Step 3.1 force all cardinalities within each class to be equal. For distinct classes Q, R with $d_{Q,R} < N$, consider the complete directed graph on the classes, with arc lengths $d_{U,V}$, and choose a shortest Q-R path having the largest number of arcs. Its total length is $d_{Q,R}$, so every arc (U, V) on this path satisfies $0 \le d_{U,V} < N$ and is considered in Step 3.2. If some arc (U, V) were not retained, the condition in Step 3.2 and the triangle inequality would imply $d_{U,V} = d_{U,T} + d_{T,V}$ for a third class T. Replacing this arc by $U \to T \to V$ would produce a shortest walk with more arcs. This walk cannot contain a positive-length cycle, which would contradict shortestness, or a zero-length cycle, whose classes would belong to the same zero-distance equivalence class. It would therefore be a path, contradicting our choice. Thus, the retained arcs reproduce every distance below N. Summing their constraints along such a path gives $C_i - C_j \le d_{Q,R} = d_{i,j}$ for every $i \in Q$ and $j \in R$. Bounds with $d_{Q,R} = N$ follow directly from $0 \le C_i \le N$. Together with equality within each class and $d_{i,j} \le \alpha_{i,j}$, these bounds imply $\boldsymbol{C} \in \mathcal{F}_0$. Conversely, every retained constraint is valid for $\mathcal{F}_0$. Hence, the reduced and original feasible regions coincide.

We now relate any equivalent representation to the same distances to prepare the minimum-cardinality argument. Consider any feasible representation $(\mathcal{A}, \boldsymbol{x}_{\mathcal{A}})$ of the reduction problem. Construct a directed graph with an arc of length $x_{i,j}$ for every $(i, j) \in \mathcal{A}$, and add auxiliary arcs of length N between all distinct nodes. These auxiliary arcs encode bounds already implied by the cardinality box and do not count toward $|\mathcal{A}|$. Let $\delta_{i,j}$ denote the resulting shortest-path distances. By the same path-summation and feasible-vector argument used above, with $C_k = \delta_{k,j}$, equivalence implies

$$\delta_{i,j} = \max_{\boldsymbol{C} \in \mathcal{F}_0} (C_i - C_j) = d_{i,j}, \qquad i, j \in \mathcal{M}.$$

We first establish a lower bound on the constraints needed within each non-singleton class. For any $Q \in \mathcal{Q}$ with $|Q| \ge 2$, its members must be mutually connected by zero-length paths because $\delta_{i,j} = \delta_{j,i} = 0$ for all $i, j \in Q$. Such paths cannot use auxiliary arcs, whose length is $N > 0$. They also cannot leave Q: any outside node on such a path would have mutual zero distances with a member of Q and hence belong to Q. Therefore, the zero-length arcs within Q form a strongly connected directed graph. Every node requires at least one outgoing arc within Q, so

$$|\mathcal{A} \cap (Q \times Q)| \ge |Q|.$$

We next establish a separate lower bound for each ordered class pair retained in Step 3.2. Recall that E denote the ordered class pairs retained in Step 3.2. Fix $(Q, R) \in E$, $i \in Q$, and $j \in R$. A shortest i-j path has length $d_{Q,R} < N$ and therefore uses no auxiliary arc. If it visited a node k in a third class T, its length would satisfy

$$d_{Q,R} \ge \delta_{i,k} + \delta_{k,j} = d_{Q,T} + d_{T,R} > d_{Q,R},$$

where the strict inequality follows from the retention condition in Step 3.2. This argument also applies when $d_{Q,R} = 0$. Thus, this path stays within $Q \cup R$ and must contain an arc from Q to R. This implies

$$|\mathcal{A} \cap (Q \times R)| \ge 1.$$

Combining these disjoint lower bounds establishes the minimum cardinality of $\mathcal{A}'$. The within-class arc sets and the arc sets associated with distinct ordered class pairs are mutually disjoint, so

$$|\mathcal{A}| \ge \sum_{\substack{Q \in \mathcal{Q} \\ |Q| \ge 2}} |Q| + |E| = |\mathcal{A}'|.$$

The equality holds because Step 3.1 selects exactly $|Q|$ arcs for each non-singleton class, and Step 3.2 selects exactly one arc for each pair in E. Since $(\mathcal{A}, \boldsymbol{x}_{\mathcal{A}})$ was arbitrary and the returned system is equivalent to $\mathcal{F}_0$, the algorithm attains the minimum cardinality.

We also verify that every retained tolerance level is tight. For each $(i, j) \in \mathcal{A}'$, the algorithm uses the tolerance $d_{i,j}$. By (C.24), the feasible vector $C_k = d_{k,j}$ attains $C_i - C_j = d_{i,j}$. Replacing this tolerance by any smaller nonnegative integer would therefore exclude this feasible vector and change the feasible region. When $d_{i,j} = 0$, no smaller nonnegative integer is admissible.

Finally, we establish the claimed running time. Step 1 and the construction of the equivalence classes and their cycles require $O(M^2)$ time. The Floyd-Warshall computation in Step 2 takes $O(M^3)$ time, and Step 3.2 examines at most $|\mathcal{Q}|^3 \leq M^3$ triples. The total running time is therefore $O(M^3)$. □

Proof of Lemma B.3: Recall that $\max_{S:|S|=C} \sum_{j\in S} h_{ij}(I_i^{(z)}) = \sum_{j\in \mathcal{B}_i^{(z)}(C)} h_{ij}(I_i^{(z)})$ and $h_{ij}(R)$ decreases in R. Therefore, $\max_{S:|S|=C} \sum_{j\in S} h_{ij}(R)$ also decreases in R and $\max_{S:|S|=C} \sum_{j\in S} h_{ij}(I_i^{(z)}) - I_i^{(z)}$ decreases in z. Since $\max_{S:|S|=C} \sum_{j\in S} h_{ij}(I_i^{(0)}) - I_i^{(0)} = \max_{S:|S|=C} \sum_{j\in S} r_j v_{ij} \geq 0$, z' always exists as the largest passing index in $\{0,\ldots,Z_i\}$.

It indicates that $R_i^*(C) \geq I_i^{(z')}$ otherwise

$$R_i^*(C) = \max_{S:|S|=C} \sum_{j\in S} h_{ij}(R_i^*(C)) > \max_{S:|S|=C} \sum_{j\in S} h_{ij}(I_i^{(z')}) \geq I_i^{(z')} > R_i^*(C)$$

and leads to a contradiction. The equality follows from Equation (A.20), and the inequalities follow from the definition of z'.

Then we will prove $R_i^*(C) \leq I_i^{(z'+1)})$. If $z' = Z_i$, then $R_i^*(C) < I_i^{(Z_i+1)} = +\infty$ because all product revenues are finite. Otherwise, assume $z' < Z_i$. If $R_i^*(C) \geq I_i^{(z'+1)}$, then

$$R_i^*(C) \geq I_i^{(z'+1)} > \max_{S:|S|=C} \sum_{j\in S} h_{ij}(I_i^{(z'+1)}) \geq \max_{S:|S|=C} \sum_{j\in S} h_{ij}(R_i^*(C)) = R_i^*(C),$$

which is a contradiction. Then we prove $R_i^*(C) \in [I_i^{(z')}, I_i^{(z'+1)})$. □